

GitHub Repositories I Can Access

john-h101
Jackyfirrest
RheaTinghe
sumeng29
JoshuaRZC
benjik2001
alexchsoh
leopold-xx
Bibi-Wangg
ChenfeiP
lylatraylor
IssacCheby
AtuOlivia
candiizy
davdpanpan
vic-safer
hyemin20

DonjhaiH
pbryann
xinpengchen29
kayantang666
oli-maynard-dev
destinyogu
larissaarreola
andreagalloro
mhigashida
Eleen-EH
markszekacs
abss123
Anchor-bkl
RoyLi01
madisonmacdonald111
victoriak2001
Jessiezhouuuu

grantzhou23
aarushmaddela
ricardo-pc
sharyali05
cylee0815
maramasic
sonyak28
evianaphm
Leoleoxu
ruihangw123
laokaoya
minikouda
evalynnaong
taylorvassar
Akanshaa18
Ziyu-Ge
eluxyl

AlexVolkov0404
quannm11

If your username is in
red, your repo is public
and needs to be changed
to private.

STAT 214 Spring 2026

Week 2

Zach Rewolinski

Any Course Questions?

Announcements

lab0 is “due” **tonight at 11:59pm**

- Push at least a `report/lab0.pdf` in a folder called `lab0`.

lab1 has been released in the [stat-214-gsi](#) repo. **Due Friday Feb 20th at 11:59pm**

- Submit in your stat-214 repo

Lab 1

How do I do lab1???

Read through:

`stat-214-gsi/lab1/instructions/lab1-instructions.pdf`

And make sure you are also familiar with:

`stat-214-gsi/disc/week1/lab-instructions.pdf`

Post on Ed or come to OH if anything doesn't make sense!

If something is mentioned in these instruction documents, it is fair game for grading. For example, the file tree structure in `lab-instructions.pdf` will be enforced in the rubric.

The Problem: Traumatic Brain Injuries & CT Scans

In class on Tuesday, a recorded talk from Dr. Aaron Kornblith was played, and he came for Q&A.

I'm hoping this experience gives you a strong amount of background knowledge regarding the dangers of traumatic brain injuries and giving CT scans to young children.

stat-214-gsi/lab1/documents/Kuppermann_2009_The-Lancet_000.pdf

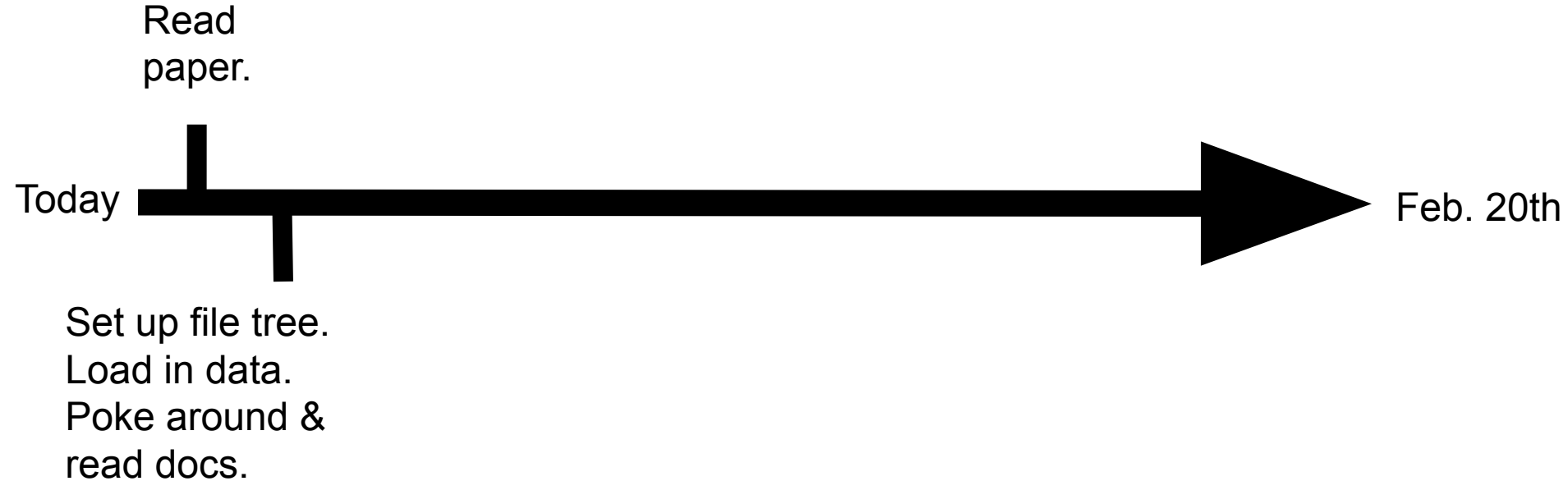
Identification of children at very low risk of clinically-important brain injuries after head trauma: a prospective cohort study

*Nathan Kuppermann, James F Holmes, Peter S Dayan, John D Hoyle, Jr, Shireen M Atabaki, Richard Holubkov, Frances M Nadel, David Monroe, Rachel M Stanley, Dominic A Borgialli, Mohamed K Badawy, Jeff E Schunk, Kimberly S Quayle, Prashant Mahajan, Richard Lichenstein, Kathleen A Lillis, Michael G Tunik, Elizabeth S Jacobs, James M Callahan, Marc H Gorelick, Todd F Glass, Lois K Lee, Michael C Bachman, Arthur Cooper, Elizabeth C Powell, Michael J Gerardi, Kraig A Melville, J Paul Muizelaar, David H Wisner, Sally Jo Zuspan, J Michael Dean, Sandra L Wootton-Gorges, for the Pediatric Emergency Care Applied Research Network (PECARN)**

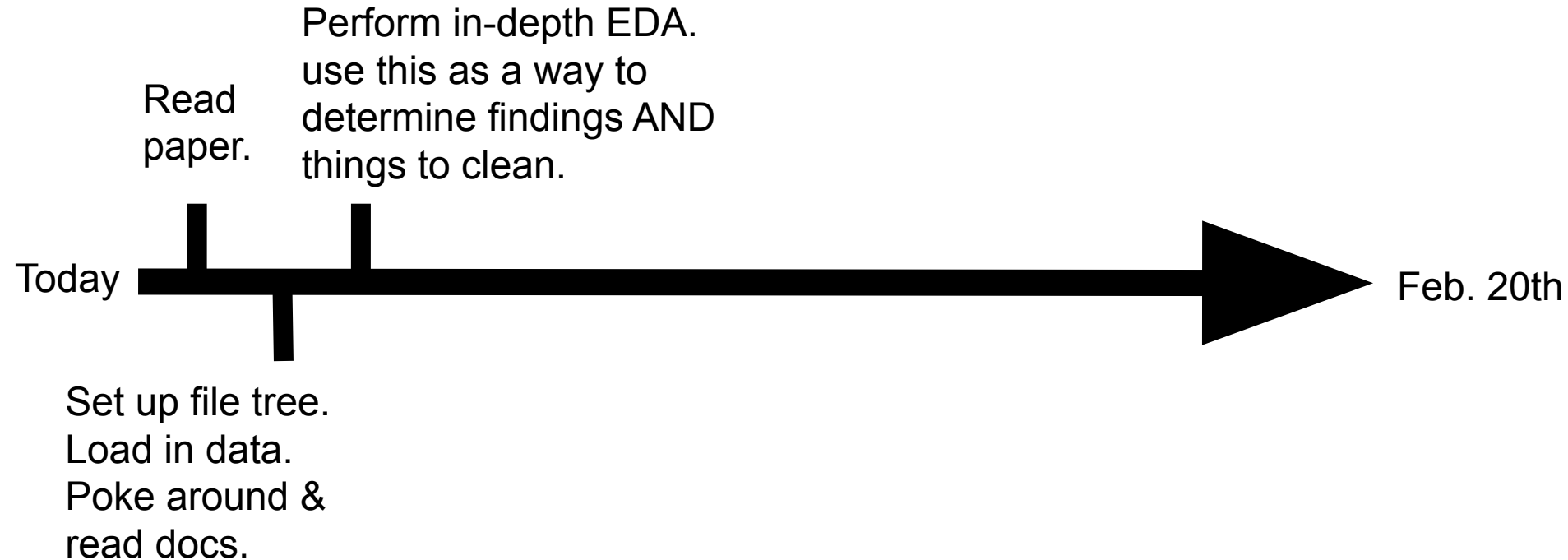
Zach's Recommended Order of Doing the Lab



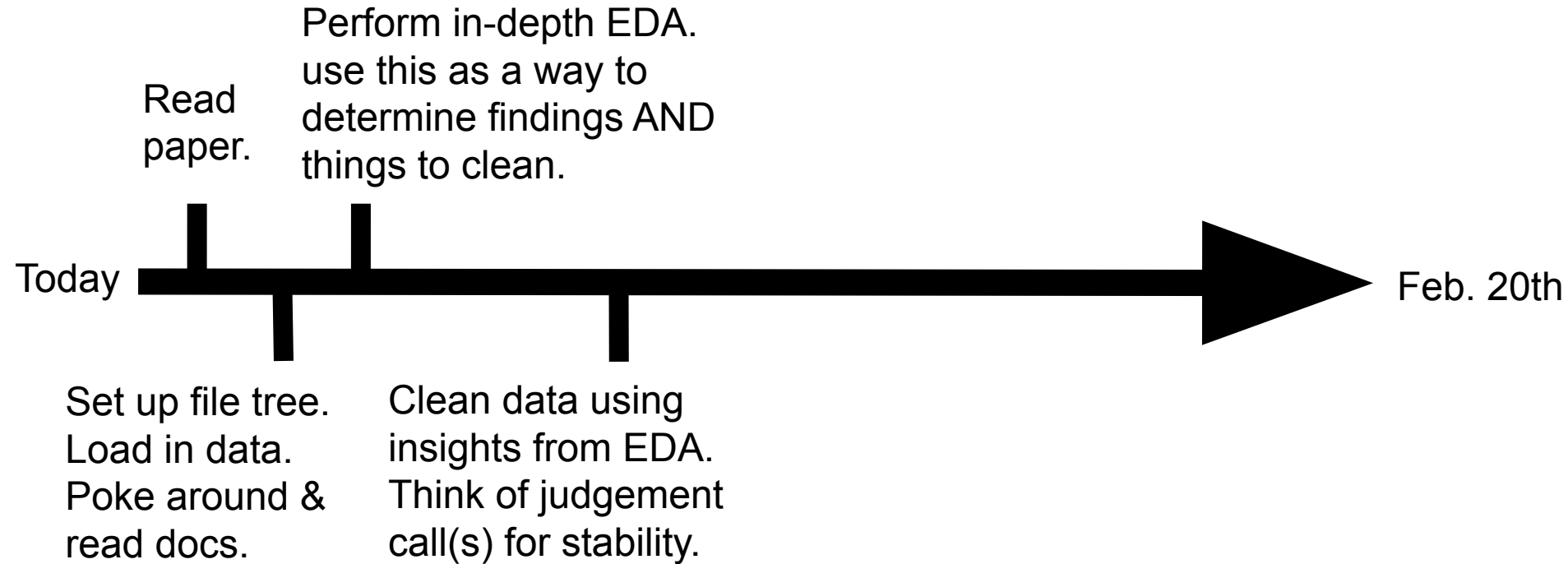
Zach's Recommended Order of Doing the Lab



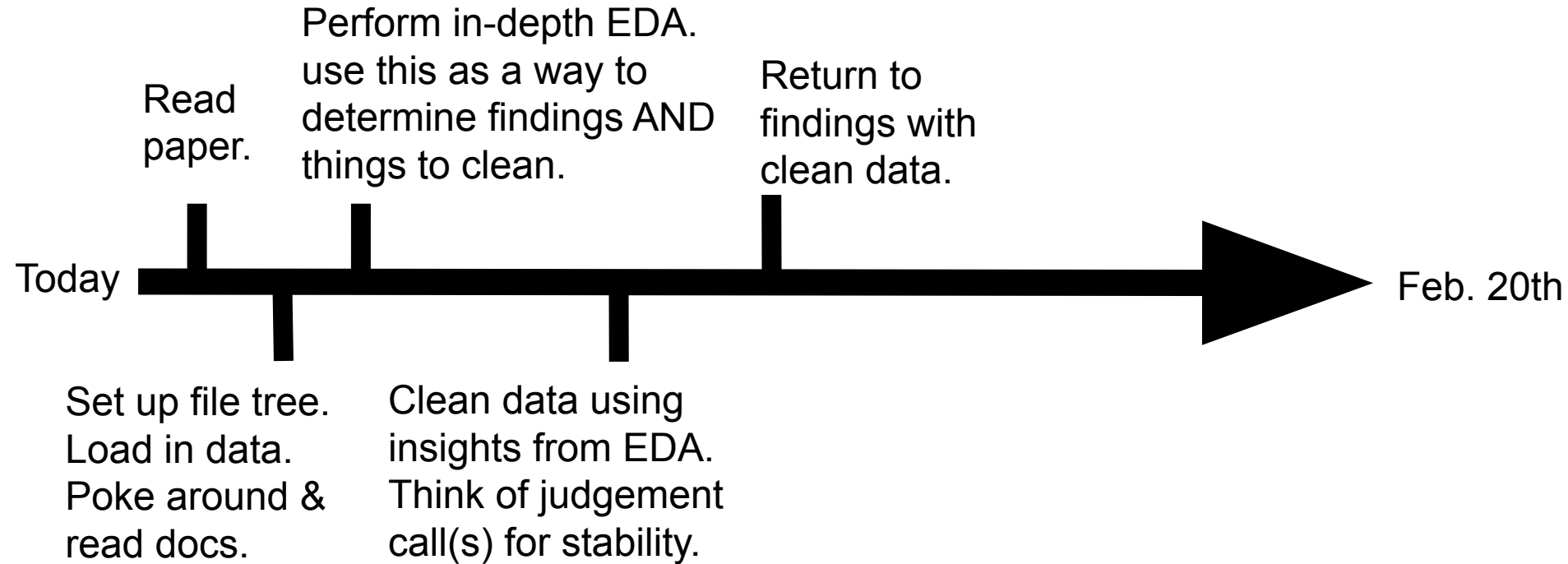
Zach's Recommended Order of Doing the Lab



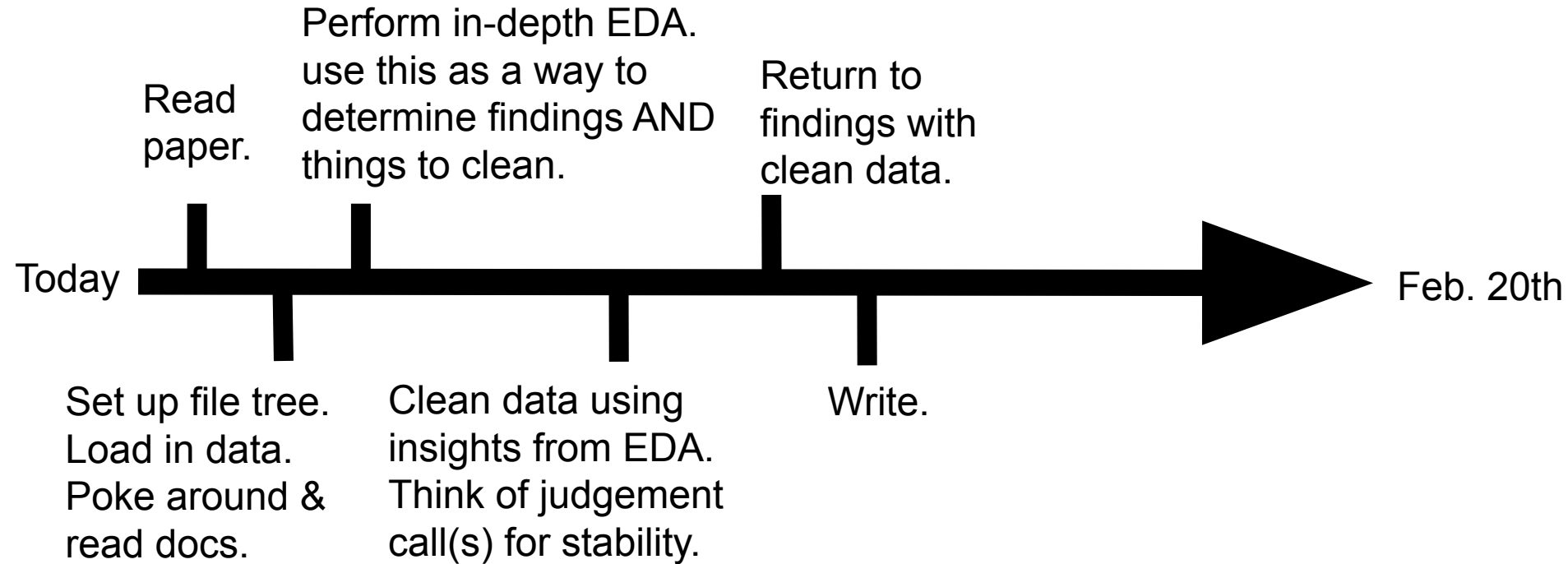
Zach's Recommended Order of Doing the Lab



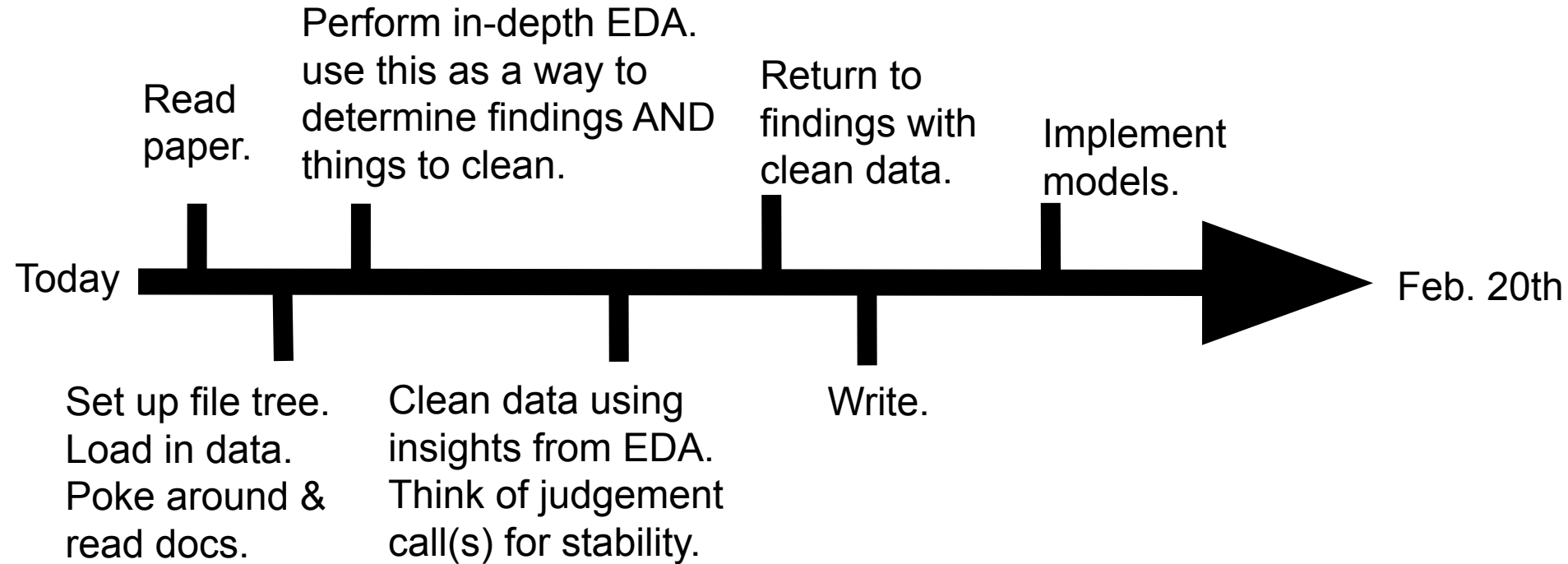
Zach's Recommended Order of Doing the Lab



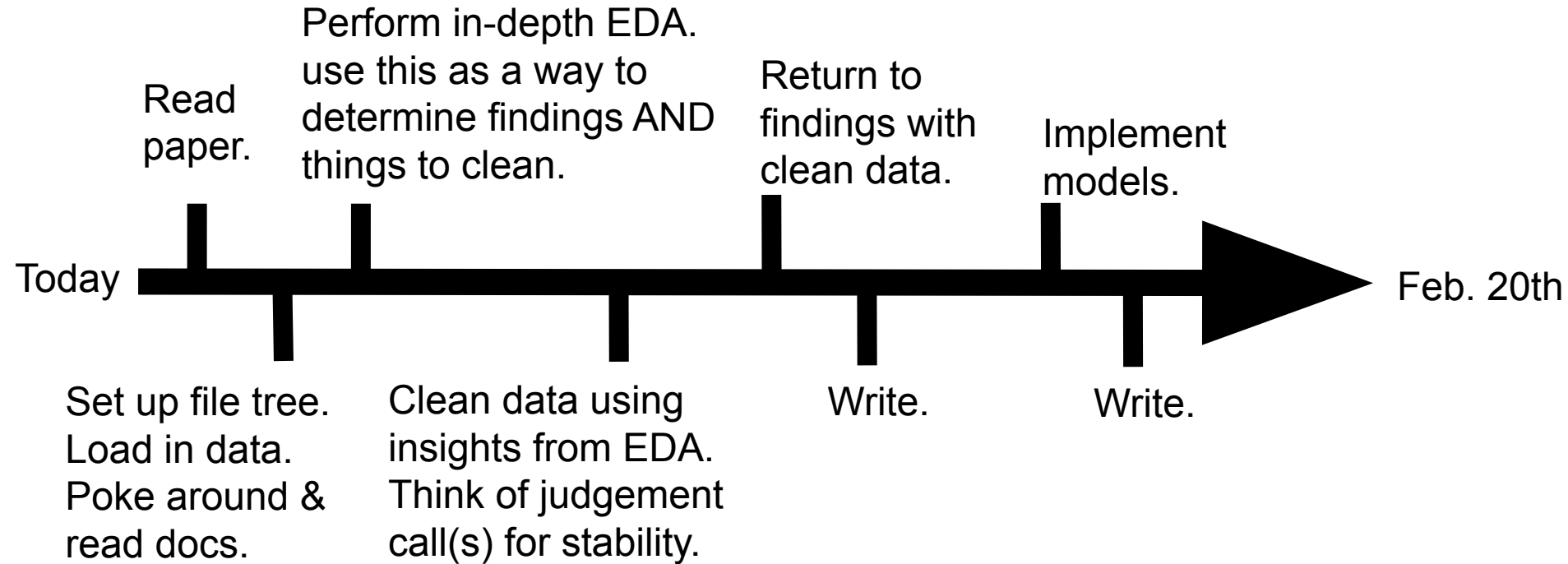
Zach's Recommended Order of Doing the Lab



Zach's Recommended Order of Doing the Lab

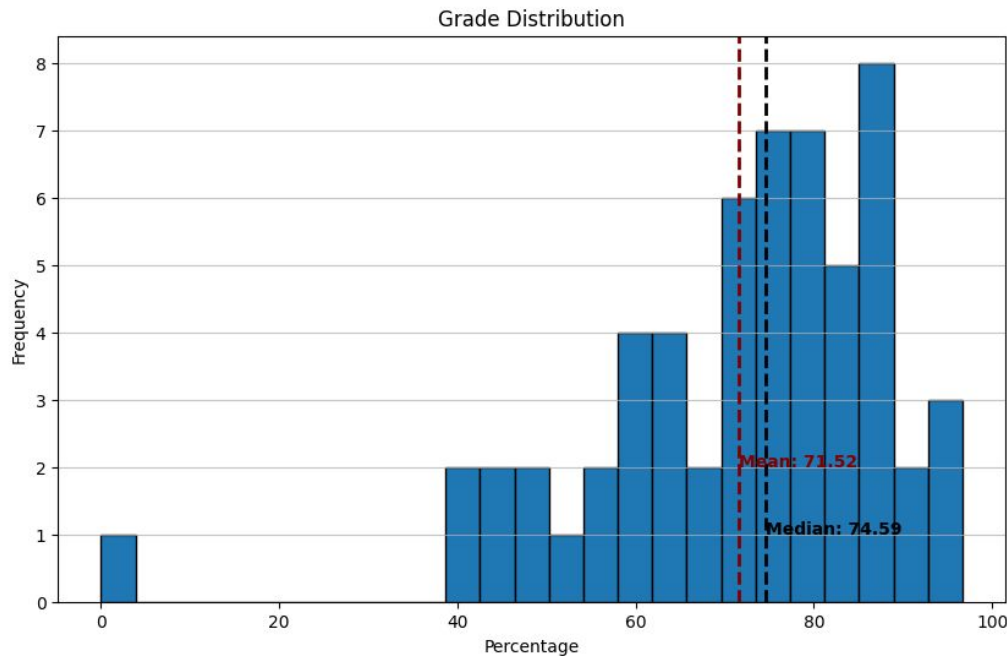


Zach's Recommended Order of Doing the Lab



It's Just Data Cleaning! That's Easy Enough, Right?

- ~20% of your Lab 1 grade will be adherence to formatting instructions.
 - Should be a significant amount of “free” points for everyone.
- Yet, this is the grade distribution from last year:
- What happened?



Reminder: I am Mainly Grading Your Report

- If you complete the best analysis ever, it won't matter much unless you showcase that extensively in your report.
- Don't wait to write the report until the end!! It is a relatively long report. Most below-average grades from last year were due to students procrastinating the writing portion of the project, and submitting incomplete reports.
- You do not need to "hit" the page limit. But you also should not be significantly shorter than the limit. Space will be needed to describe your reasoning, and figures will take up lots of space.

What Counts as an “Interesting Finding”

Together, the three “findings” are a large part of your grade. I advise you to spend a significant chunk of time doing deep EDA to dig into the quirks of the data.

As to not give away any secret info, it is easier for me to discuss what is not an “interesting finding”.

- Correlation matrices/heatmaps/etc.
- Something ridiculously obvious. Example: patients with incredibly low GCS scores have more ciTBIs. Shocking!
- General rule of thumb: if it takes less than five minutes to find, it is likely to score low.

What Does a High-Quality Figure Look Like?

A good figure should be:

- Simple
- Informative
- Honest
- Efficient
- Aesthetic

Things to consider:

- Audience: Who is the figure intended for?
- Convention: How do people usually visualize similar things?
- Purpose: What should the figure accomplish? Is it useful?
- Medium: How will the viewer see the figure? How big will the figure be?

What Does a High-Quality Figure Look Like?

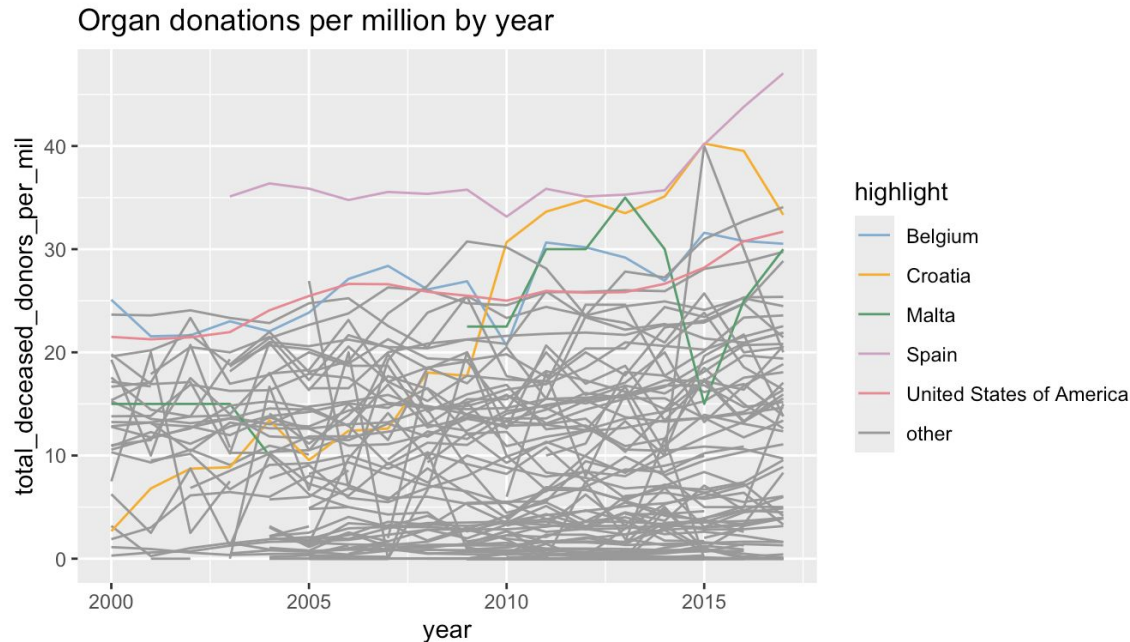
Things to avoid:

- Pure `matplotlib` defaults (will get points taken off)
- Counterintuitive color schemes
- Raw column names
- Overcrowded figures

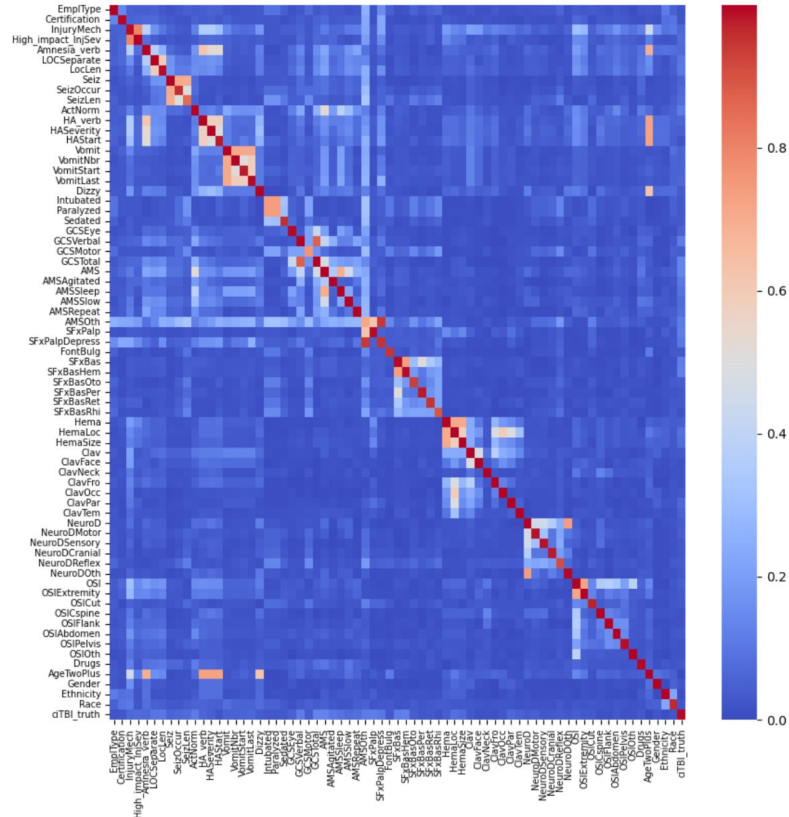
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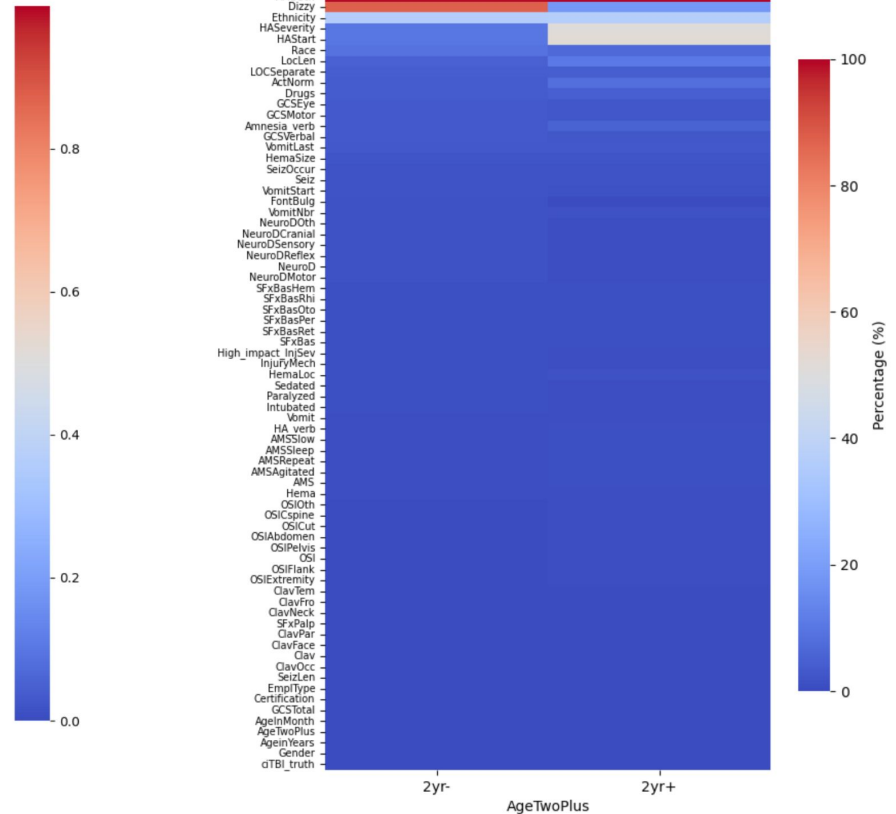
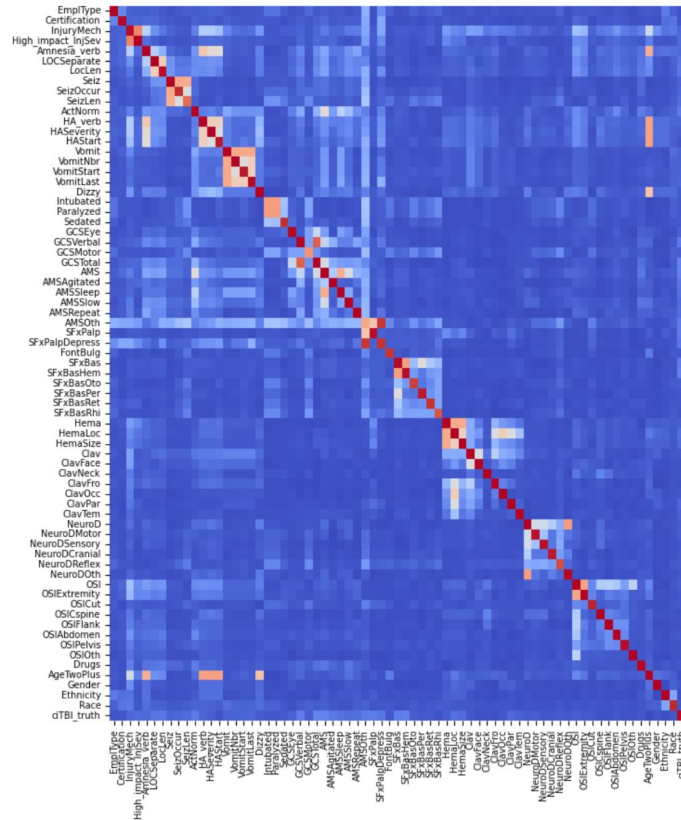
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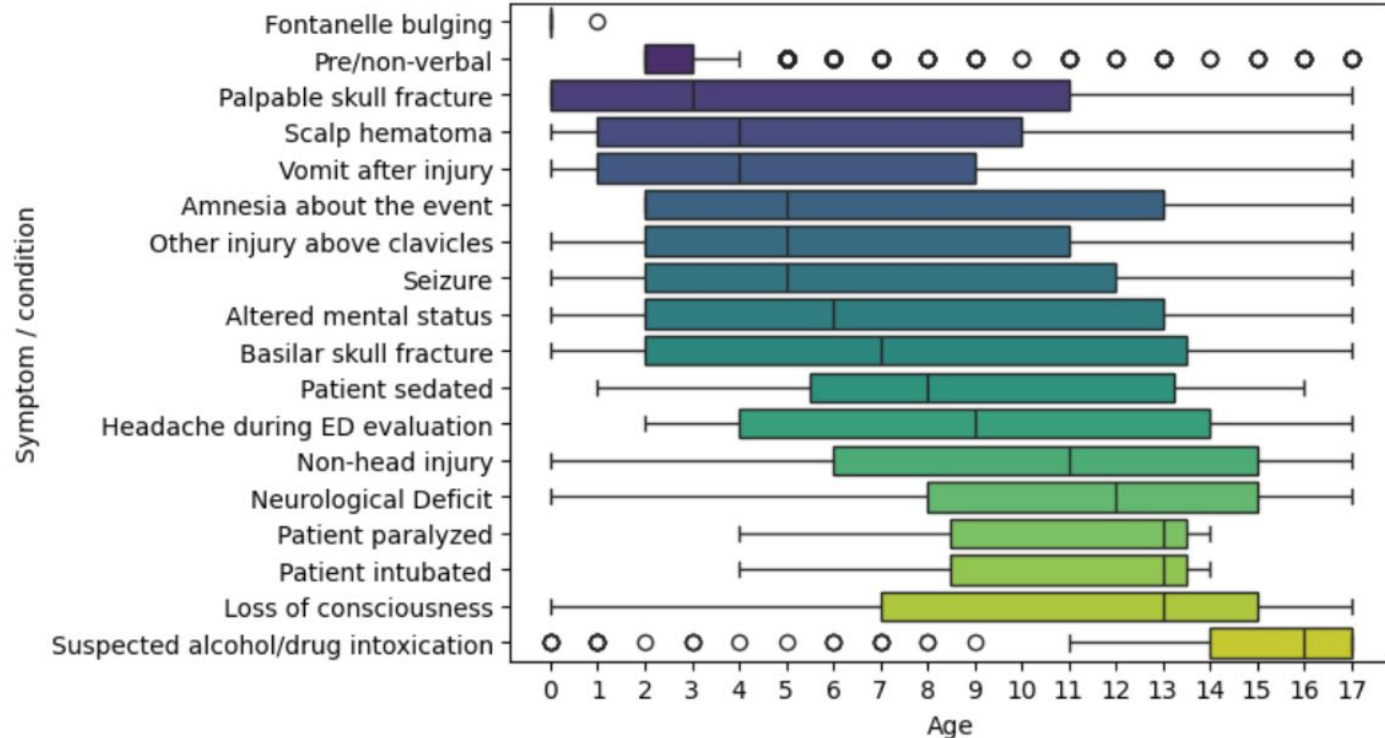
What Does a High-Quality Figure Look Like?



What Does a High-Quality Figure Look Like?



What Does a High-Quality Figure Look Like?



For This Project, What is “Clean” Data?

This data is... very, very dirty. Without giving an exhaustive list (can't make your lives too easy), here are some examples:

- There are data entry errors
- There are different types of missingness
- There are dependencies between columns (do they agree???)
- ... and much more

I should be able to tell how deeply you dove into data cleaning by seeing the problems that you describe in your report.

Data cleaning in general

What does “clean” data look like?

Data cleaning in general

What does “clean” data look like?

- No invalid/inconsistent values. This is based on domain knowledge.
 - What do you do with rows containing invalid values? Remove the whole row? Impute the values? Change the values to missing?
- Missing values are appropriately encoded.
 - e.g. `np.nan` instead of `-1`.
 - It is okay to keep missing values in your clean data as long as your downstream models appropriately address them.
- Nice data format. Ideally a tabular format, but this isn't possible with all datasets.
- Consistent and clear naming scheme for columns and categorical values
- Appropriate datatypes are used
 - (e.g. categorical data shouldn't be stored as float, booleans should be bool, etc.)
- Every observational unit should only appear once in the data

See:

`stat-214-gsi/discussion/week2/pandas.ipynb`

for some pandas examples and practice, if you would like.